SS-P-1540B
September 25, 1973
SUPERSEDING
Fed. Spec. SS-P-1540A
December 12, 1969

FEDERAL SPECIFICATION

PIPE, BITUMINIZED FIBER (AND FITTINGS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers homogeneous and laminated bituminized fiber pipes of solid wall and perforated wall construction, and the fittings and couplings for these pipes (see 6.1).

1.2 Classification.

1.2.1 Type, class, style, and size. Pipe furnished under this specification shall be of the following types, classes, styles, and sizes, as specified (see 6.2):

Type I - Homogeneous wall structure.

Class 1 - Non-perforated wall.

Size - 2, 3, 4, 5, 6, and 8 inches I.D.

Class 2 - Perforated wall.

Style A - General drainage.

Size - 2, 3, 4, 5, and 6 inches I.D.

Style B - Septic tank disposal fields.

Size - 4 inches I.D.

Style C - Airport and roadwork drainage.

Size - 2, 3, 4, 5, 6, and 8 inches I.D.

Type II - Laminated wall structure.

Class 1 - Non-perforated wall.

Size - 2, 3, 4, 5, 6, and 8 inches I.D.

Class 2 - Perforated wall.

Style A - Agricultural drainage.

Size - 3, 4, 5, 6, 8, 10, 12, and 15 inches I.D.

Style B - Septic tank disposal fields.

Size - 3 and 4 inches I.D.

Style C - Airport and roadwork drainage.

Size - 3, 4, 5, 6, 8, 10, 12, and 15 inches I.D.

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1.2.2 Couplings and pipe fittings. Couplings and pipe fittings such as bends tees and wyes shall be homogeneous or laminated construction.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D1861 - Specification for Homogeneous Bituminized Fiber Drain and Sewer Pipe.
D1862 - Specification for Laminated-Wall Bituminized Fiber Drain and Sewer Pipe.
D2311 - Specification for Homogeneous Perforated Bituminized Fiber Pipe for General Drainage.
D2312 - Specification for Homogeneous Perforated Bituminized Fiber Pipe for Septic Tank Disposal Fields.
D2313 - Specification for Laminated-Wall Perforated Bituminized Fiber Pipe for Septic Tank Disposal Fields.
D2314 - Testing Homogeneous Bituminized Fiber Pipe.
D2315 - Physical Testing of Laminated-Wall Bituminized Fiber Pipe.
D2417 - Specification for Laminated-Wall Bituminized Fiber Perforated Pipe for Agricultural, Land, and General Drainage.
D2418 - Specification for Laminated-Wall Bituminized Fiber Perforated Pipe for Airport and Roadwork Drainage.
D2818 - Specifications for Homogeneous Perforated Bituminous Fiber Pipe Systems for Highway, Airport, and Similar Drainage.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General. Bituminized fiber pipe shall be either Type I, homogeneous wall structure, or Type II, laminated wall structure. The pipe shall be either perforated wall or nonperforated wall, as applicable, and shall be as specified herein.

3.1.1 Material. Pipe and couplings shall be composed of a bituminous compound reinforced with a fibrous structure. The fibrous material shall be thoroughly impregnated. Couplings and fittings of composition other than pitch-fiber may be used provided they have equal or better chemical and physical characteristics.

3.1.2 Physical, chemical, and dimensional. Physical, chemical, and dimensional requirements for pipe and couplings shall conform to the ASTM standard specified ~~for the applicable class or class and style pipe~~.

3.2 Type I, Homogeneous wall structure.

3.2.1 Class 1, nonperforated wall pipe. Class 1 pipe shall conform to the requirements of ASTM D1861.

3.2.2 Class 2, perforated wall pipe.

3.2.2.1 Style A, general drainage. Style A pipe shall conform to the requirements of ASTM D2311.

3.2.2.2 Style B, septic tank disposal fields. Style B pipe shall conform to the requirements of ASTM D2312.

3.2.2.3 Style C, airport and roadwork drainage. Style C pipe shall conform to the requirements of ASTM D2818.

3.2.3 Construction. Type I pipe, couplings, and fittings shall be constructed of a dense homogeneous bituminized fiber wall structure, and shall be tested in accordance with 4.5.1. Couplings and fittings of composition other than pitch-fiber may be used provided they have equal or better chemical and physical characteristics.

3.3 Type II, Laminated wall structure.

3.3.1 Class 1, nonperforated wall pipe. Class 1 pipe shall conform to the requirements of ASTM D1862.

3.3.2 Class 2, perforated wall pipe.

3.3.2.1 Style A, agricultural drainage. Style A pipe shall conform to the requirements of ASTM D2417.

3.3.2.2 Style B, septic tank disposal fields. Style B pipe shall conform to the requirements of ASTM D2313.

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3.3.2.3 Style C, airport and roadwork drainage. Style C pipe shall conform to the requirements of ASTM D2418.

3.3.3 Construction. Type II pipe, couplings, and fittings shall be constructed of a bituminized compound reinforced with a multi-ply laminated convolute or spiral fibrous structure, and shall be tested in accordance with 4.5.2. Couplings and fittings of composition other than pitch-fiber may be used provided they have equal or better chemical and physical characteristics.

3.4 Standard lengths. Specified standard lengths (see 6.2) of bituminized fiber pipe shall conform to the requirements of table I.

TABLE I.
Standard lengths of bituminized fiber pipe

Type	Pipe size, inside diameter, inches	Standard lengths, feet
I	2, 3, 4, 5, 6, and 8	5, 8, or 10
II	2, 3, 4, and 5	5, 8, or 10
II	6, 8, 10, 12, and 15	5, 8, 10, or 20

3.5 Couplings. Unless otherwise specified, (see 6.2), one coupling of the same type as the pipe specified (see 6.2) shall be furnished with each length of pipe and with each fitting. Couplings and fittings of composition other than pitch-fiber may be used provided they have equal or better chemical or physical characteristics.

3.5.1 Straight couplings with tapered inside wall.

3.5.1.1 Homogeneous bituminous fiber couplings. Couplings shall be in accordance with the following applicable ASTM designations: D1861, D2311, or D2312.

3.5.1.2 Laminated bituminous fiber couplings. Couplings shall be in accordance with the following applicable ASTM designations: D1862, D2313, D2417, or D2418.

3.5.2 Couplings, 5 degree angle.

3.5.2.1 Homogeneous bituminous fiber couplings. Couplings for drain and sewer pipe shall be in accordance with ASTM D1861.

3.5.2.2 Laminated bituminous fiber couplings. Couplings shall be in accordance with the following applicable ASTM desginations: D1862, D2417, or D2418.

3.5.3 Split collars.

3.5.3.1 Homogeneous bituminous fiber collars. Collars for perforated pipe used for highway, airport and similar drainage shall be in accordance with ASTM D2311, D2312, or 2818.

3.5.3.2 Laminated bituminous fiber collars. Collars shall be in accordance with the following applicable ASTM designations: D2417, D2418, or D2313.

3.6 Pipe fittings. Fittings shall be fabricated from pipe, or molded at the manufacturers option. Couplings and split collars shall be of applicable pipe sizes, as specified (see 6.2). Couplings and fittings of composition other than pitch-fiber may be used provided they have equal or better chemical and physical characteristics.

3.6.1 Bends.

3.6.1.1 Homogeneous bituminous fiber bends. Homogeneous bituminous fiber bends shall be in accordance with ASTM D1861. Bends shall be 45 or 90 degrees as specified (see 6.2). Radius and size of bends shall be as follows, and as specified (see 6.2):

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Pipe size Inches	Radius of bend Inches	Bend Degrees
2	9.5, 18, 24, 36	45, 90
3	13, 24, 36	45, 90
4	16, 36	45, 90
5	24, 36	45, 90
6	36	45, 90
8	48	45

3.6.1.2 Laminated bituminous fiber bends. Laminated bituminous fiber bends shall be in accordance with ASTM D1862, D2313, D2417 or D2418. Bends shall be 22-1/2, 45 or 90 degrees as specified (see 6.2). Radius shall be fabricated at the manufacturers option of reinforced, multi-ply laminated convolute or spiral fabricated structure or a radius fabricated of pipe segments, angularly form, joints cemented with a composition of pitch fiber to provide physical characteristics equal or better than the parent material to form bends 22-1/2, 45 or 90 degrees. Radius degrees and size of bends shall be as follows and as specified (see 6.2):

Pipe size Inches	Radius of bend Inches	Bend Degrees
2	9.5	45 or 90
3	10.8	45 or 90
4	13.25	22-1/2, 45 or 90
6	18.12	45 or 90
8	23.0	45 or 90

3.6.2 Adapters. One end shall enclose the spigot end of cast iron pipe, clay pipe, plastic pipe, or pipe of any other material, as specified (see 6.2). There shall be sufficient annular space to pack the joint with jointing material. The opposite end shall have an external or internal tapered joint, as specified (see 6.2). When specified (see 6.2), the opposite end shall have internal threads for attachment to threaded steel pipe. Adapter threads shall conform to NBS H28.

3.6.3 Reducers. Reducers shall have internal taper joints at each end.

3.6.3.1 Homogeneous bituminous fiber reducers shall be of the following sizes, as specified (see 6.2):

Large end Inches	Small end Inches
8	6, 5, 4, 3, 2
6	5, 4, 3, 2
5	4, 3, 2
4	3, 2
3	2

3.6.3.2 Laminated bituminous fiber reducers. Reducers shall be of the following sizes, as specified (see 6.2):

Large end Inches	Small end Inches
8	6, 4, 2
5	4, 2
3	2

3.6.4 Tees and laterals. The end run of the branch shall have internal or external taper joints, as specified (see 6.2). Size of tees and laterals, and shape of laterals including wyes shall be as specified (see 6.2).

3.6.5 Plugs and other fittings. Type of fitting, size, and when material is required shall be as specified (see 6.2).

3.7 Interchangeability. All joints of the same type, class, style, and size shall be completely interchangeable.

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3.8 Interior surfaces. Interior surfaces of pipe and fittings shall be smooth, and free from obstructions and rough or flaky areas. Examination shall be in accordance with 4.4.

3.9 Marking for identification. Pipe and fittings shall be clearly and legibly marked in accordance with the requirements of MIL-STD-130, and shall be examined in accordance with 4.4.

3.10 Workmanship. Pipe, fittings, and couplings furnished under this specification shall be of uniform quality and fabricated in accordance with the best commercial practices.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all pipe of the same type, class, or style, and all fittings and couplings furnished for the type of pipe specified, offered for delivery at the same time.

4.2.2 Sampling for examination and tests. Sampling for visual examination and inspection of preparation for delivery shall be in accordance with level II in MIL-STD-105. Sampling for tests shall be in accordance with level S-2 in MIL-STD-105.

4.3 Acceptable quality levels. The Acceptable Quality Level (AQL) for visual and dimensional examination inspection of preparation for delivery shall be 2.5 percent defective for major defects. The (AQL) for tests shall be 4.0 percent defective.

4.4 Visual and dimensional examination. Pipe, couplings, and fittings shall be examined and measured for the defects listed in table II.

TABLE II. Classification of defects

Defects	Major	Minor
Type, class, style, or size not as specified.	X	
Length not as specified.	X	
Fittings not furnished as specified.	X	
Couplings not furnished as specified.	X	
Structural defects: pipe, fittings and couplings		
Joint angles and depth not as specified.	X	
Wall thickness not as specified.	X	
Inside diameter not as specified.	X	
Seams, cracks, or splits.	X	
Lamination adherence by mechanical means.	X	
Visible lamination separation.	X	
Joints not interchangable.	X	
Interior surfaces not as specified.	X	
Marking incorrect, illegible, or missing.		X
Workmanship not as speciried.	X	

4.5 Test methods.

4.5.1 Type I. Test pipe, fittings, and couplings in accordance with ASTM D2314.

4.5.2 Type II. Test pipe, fittings, and couplings in accordance with ASTM D2315.

4.6 Inspection of preparation for delivery.

4.6.1 Preproduction pack inspection. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection.

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4.6.2.1 Unit of product. For the purpose of inspection, a completed pack of any one level prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with level II in MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101.	Materials, methods, or containers not as specified for level A. Each incorrect material, method, or container shall constitute one defect.
102.	Strapping not zinc-coated for level A.
103.	Marking illegible, incorrect, or incomplete for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Unless otherwise specified in the contract or order, the pipe and fittings shall be packaged and packed for domestic or export shipment in accordance with the manufacturers recommendations, which will insure carrier acceptance and safe delivery at the lowest rates to the ultimate destination. Shipping containers shall be in accordance with the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.2.1 or 5.2.2 as specified (see 6.2).

5.2.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Bituminized fiber pipe is used in non-pressure conductor and drain system. Both type I (homogeneous wall structure) pipe and type II (laminated wall structure) pipe are fabricated with non-perforated walls and perforated walls. The non-perforated wall pipe (class 1) may be used for house connections to sewers and septic tanks, industrial waste and storm water drainage, leaders, protective covering for other underground pipe, downspouts, salt water disposal in oil well operations, sleeves in concrete walls, weep holes and other uses as indicated by the pipe's characteristics. Perforated wall pipe (class 2) systems are used as under drains with infiltration in subsurface drainage for agricultural lands, airports, roads, foundations and the like, as well as for the disposal of septic tank effluent with exfiltration (see 1.1).

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style (if appropriate), and size of pipe (see 1.2.1).
(c) Desired lengths (see 3.4) and number or total linear feet of each length.
(d) If couplings are not to be included (see 3.5).
(e) Nomenclature and number of fittings (see 3.6).
(f) Size of lot if different from 4.2.1.
(g) When specified time frame required for submission of the preproduction pack (see 6.3).
(h) Packaging and packing required (see 5.2).
(i) Size of couplings, 5 degree couplings and collars (see 3.5.2 and 3.5.3).
(j) Bends, degree of bend, and size (see 3.6.4.1 and 3.6.4.2).
(k) Adapters, type of end joint, and type of jointing pipe (see 3.6.5).
(l) Reducers, size (see 3.6.6.1 and 3.6.6.2).
(m) Tees and lateral, size, joint, and shape for laterals (see 3.6.7).
(n) Plugs and other fittings, type of fitting, size, and material when specified (see 3.6.8).

6.3 Preproduction pack. Any changes or deviations of production packs from the approval preproduction pack will be subject to the approval of the contracting officer. Approval of the production pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the pipe and fittings in accordance with this specification (see 6.3).

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6.4 Cross reference data. Cross reference of classification changes between this specification and the preceding Fed. Spec. SS-P-1540A dated December 12, 1969 are as follows:

SS-P-1540A	SS-P-1540B
None	Type I, class 2, style C

MILITARY CUSTODIANS:

Army - ME
Air Force - 84

User activity:

Army - CE

Military Coordinating Activity:

Navy - YD

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE-NBS
GSA-FSS
USDA-AFS

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